



Original research article

Adaptive image steganography based on transform domain via genetic algorithm

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ABSTRACT

This paper presents a novel approach for data hiding in frequency domain with the use of genetic algorithm. At first, cover images are mapped to a proper frequency domain using the concepts of adaptive wavelet transform and genetic algorithm. In the obtained space, using a model based on Kieu and Chang, encrypted information will be embedded in the frequency coefficients that represent edges of the image in spatial domain. So the cover image will change the least and have the most compatibility with human visual system. Simulation results show that our proposed method outperforms recently published works in terms of PSNR and PSPNR factors.

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1. Introduction

With the rapid increase of internet users and their joint use of public communication channels for transmitting digital data, data security has become an inseparable part of networks. To ensure data safety, various software level encryption methods such as DES [1] or RSA [2] can be used to transfer information in shared channels. Also there are some optical methods which encrypt images in a way that it will not be easy to retrieve original image without keys [3–5]. Nature of cryptographic algorithms which make information incomprehensible to unauthorized receptors may make some attackers more curious to get activated and break the encryption keys. If Attackers cannot retrieve the original information they may manipulate transmitted stream bits so authorized receivers cannot retrieve original secret information neither. Therefore, more intelligent data hiding methods such as steganography and watermarking have emerged in the field of data transmission in order to ensure data security. Steganography and watermarking algorithms have almost similar structures but totally different aims. In these approaches, secret information are hidden in a cover such as audio, video or image in a way that the attacker who is monitoring cover media do not suspect existence of information. Use of image as cover media is attracting a lot of interest in recent studies because of image abundance on the Internet. Our proposed method is also based on this type of cover media.

In contrast to the aforementioned algorithms, some algorithms, called steganalysis, try to discover the secret hidden information in cover objects. These algorithms usually try to find weaknesses of steganography algorithms and aim to discover existence of secret information in the cover by studying cover's structural and basic features which are very changeable. So steganography algorithms should be robust against steganalysis methods.

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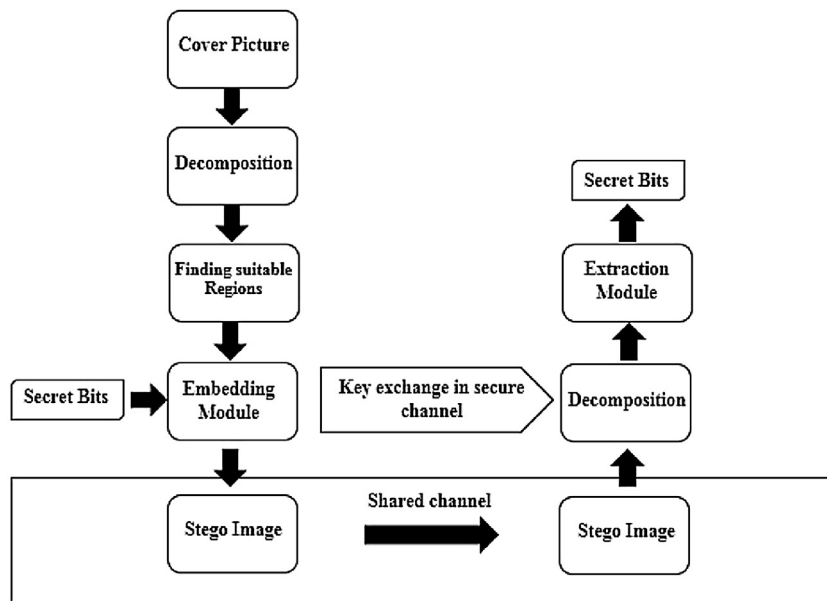


Fig. 1. Block diagram of common steganography methods in frequency domain.

Data hiding algorithms are usually implemented in both frequency and spatial domains which both have strengths and weaknesses. Some of the most important parameters that every data hiding method should have are: imperceptibility, robustness against attacks (such as compression and histogram equalization processing), high data-carrying capacity and security.

One of the most popular data hiding methods which is used in both spatial and frequency domains is LSB-based approaches [6,7]. These approaches can easily be implemented and with the use of them, large volume of information can be hidden in cover images. In [8] a method based on Genetic Algorithm is proposed to increase the quality of stego image in spatial domain. In [9] bit layers selection is optimized using genetic algorithm and also the proper offset for start point of information embedding is calculated. In most of these methods information is put all over the image randomly and edge pixels are not considered specifically. However in some methods, at first some an edge detection method like [10] is applied on the cover image and then secret data is imported in the obtained edges. In another paper, Weiqi Luo et al. [11] proposed a method in which edge pixels are chosen for information transmission in a way that stego image quality is preserved well enough that human vision system would not be able to detect the hidden information in the image. With the popularity growth of artificial intelligence optimization algorithms use in image processing, best pixels for data embedding in spatial domain can be selected with a PSO optimizing algorithm, as in [12] in a way that vision quality of cover image will be conserved as well.

Frequency based approaches use frequency transformations such as cosine, furrier, wavelet transformation for information transmission. Fig. 1 shows the overall steganography block diagram in frequency domain; this method is used as the base of our proposed method. As it is illustrated, in frequency based approaches, at first the cover image is mapped to a specific frequency space. Then proper coefficients are chosen to carry secret information; these coefficients are usually in the edges of the image or those pixels which form details. Information on how to find the coefficients is exchanged as a shared key between transmitter and receiver so that the receiver can operate in sync with the transmitter. Then the frequency coefficients which are tailored for conveying message bits will be modulated and the recipient can extract the information with demodulating them. The modulated cover image, named stego image, will be transmitted in a public channel. This channel is capable of being monitored or changed by other receptors. When the stego image is arrived, receiver will be able to extract the desired information using the shared key.

In [13], a method is proposed in which quantized coefficients of the cosine transformation is changed in order to increase the capacity of transferable information and improve the image quality. In [14], a method based on integer wavelet transformation is presented that acts based on MSBs of bit layers in the frequency domain. Another high capacity approach is proposed in [15] that uses wavelet fusion technique for color images.

In [16], the cover image is divided to sub-images, using a frequency matrix and then they are mapped to an unknown frequency domain. After information embedding, inverse transform is applied and the stego image is obtained. This method doesn't use the common frequency methods and tries to map the image to an unknown space by using a matrix. This idea is the base of our proposed method, but the unknown space will not be static and will be selected and mapped in a better way with respect to the secret information and the cover image.

Another challenge which is studied in the previous suggested methods is that receiver side is unable to estimate exact locations of information carrier coefficients. This estimation should be based on the information embedding method. A lot

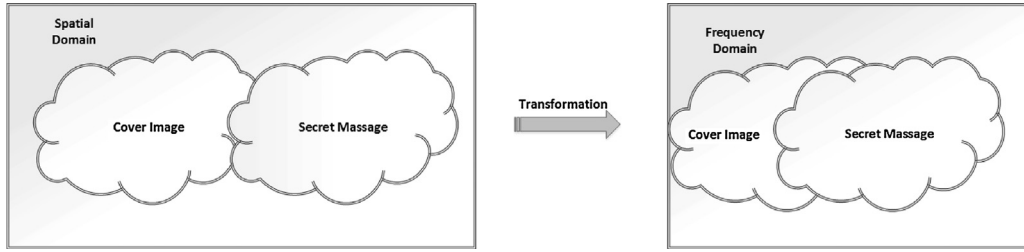


Fig. 2. Making encrypted information and cover image more alike after transformation.

of recent studies try to solve this problem and reduce the error which is caused by the transmitter and receiver not being synchronized [17].

In this paper a novel algorithm is proposed that tries to find a frequency mapping which can match the secret data with the LSB of frequency coefficients of the cover image in the best way possible; Genetic Algorithm (GA) is employed to find optimal frequency transformation parameters. Rest of the paper is as follows. In Section 2, the proposed algorithm is explained in details. Simulation results are presented in section three in which they are compared with other related algorithms and at last the conclusion is given in Section 4.

2. Proposed method

In this section adaptive wavelet transform is introduced and then the proposed method based on genetic algorithm is explained.

2.1. Adaptive wavelet transform

In a data hiding method in frequency domain, researchers usually select a specific frequency space to implement their algorithm. But in such a way, with the release of algorithm details, attacker would know the frequency space in which he/she can find the secret information. And also, it is quite possible that the chosen frequency space is not the best space adapted with the cover image and secret information. So it was suggested that the frequency space should be chosen in a way that mapping the cover image to that space, maximizes similarity of secret information and obtained frequency values (Fig. 2). As the result, after embedding data in the frequency coefficients and returning to the spatial domain, cover image would change the least.

In this paper, we have used wavelet transform and tried to estimate mother wavelet parameters in a way that our goal which is maximizing cover image and stego image resemblance is achieved. Due to (1), a two dimensional linear frequency modulation (LFM) [18] Mother Wavelet matrix is used in this paper.

$$\psi(x, y) = \sum_{p=1}^N B_p h_p(x, y) \quad (1)$$

where B_p is the coefficient of linear combination of base functions and $h_p(x, y)$ is a weighted Gaussian function as followed:

$$h_p(x, y) = g\left(\frac{\sqrt{(x-x_0)^2 + (y-y_0)^2}}{S_p}\right) \exp(j2\pi f_p \sqrt{(x-x_0)^2 + (y-y_0)^2} + j2\pi l_p((x-x_0)^2 + (y-y_0)^2)) \quad (2)$$

$$g(x, y) = \pi^{-14} \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2}\right) \quad (3)$$

in which S_p is the standard deviation of Gaussian function, x_0 and y_0 are the translation values and f_p is the coefficient variant in terms of x and y , l_p is the coefficient variant in terms of x^2 and y^2 . f_p and l_p are the parameters of the mother wavelet. For gaining desired parameters of mother wavelet, average value of $\psi_{(x,y)}$ should be put zero. As the result, after mapping the image from the frequency domain to spatial domain, edges of the image will get higher values in contrast to the uniform parts of the image. Various wavelets with different forming parameters are shown in Fig. 3; each of them yields its own unique frequency domain.

2.2. Proposed method based on genetic algorithm

In order to match the secret data with the cover image in the frequency domain and extract proper parameters for mother wavelet transform, genetic algorithm is used in this study. The desired parameters are set as the values of the genes

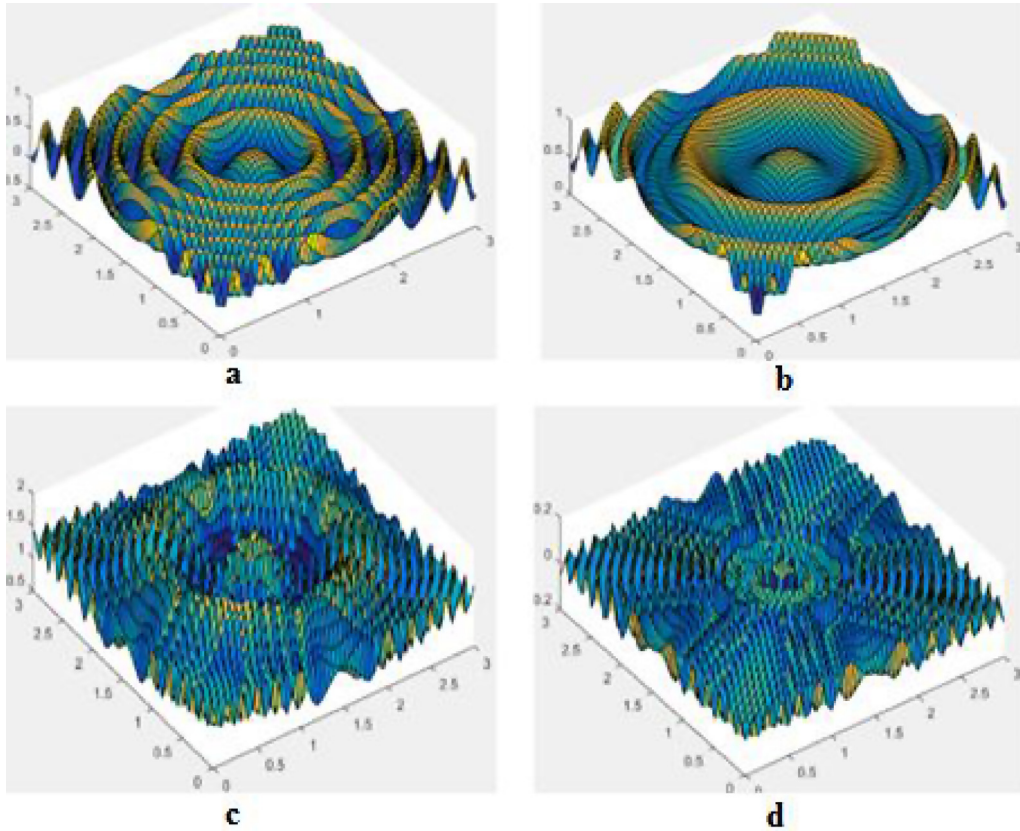


Fig. 3. Different mother wavelets generated as results of few changes in their control parameters.

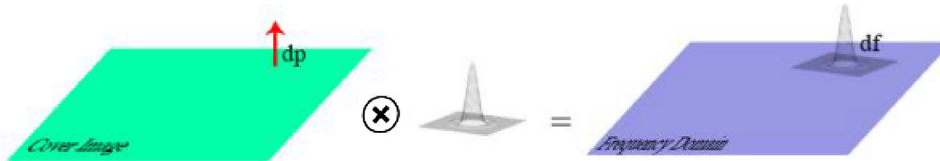


Fig. 4. Differential effect of one gray level in spatial domain on the frequency coefficients.

in chromosomes of the initial population in genetic algorithm. After extracting transform function parameters, 3×3 mother wavelet matrixes should be calculated. Then (4) is used for mapping the image from spatial domain to the frequency domain.

$$F(u, v) = \sum_{i=-1}^{i=1} \sum_{j=-1}^{j=1} \psi(x_0 + i, y_0 + j) \times I(u + i, v + j) \quad (4)$$

in which F is a frequency matrix and I is the image brightness intensity in spatial domain. Thus, for every chromosome of the population, a frequency matrix is obtained. Gained frequency coefficients are float values. Changing the frequency values just based on encrypted data with no specific consideration and then retrieving it to spatial space will produce non integer numbers. Non integer numbers cannot be used for brightness intensity of the pixels and rounding them will result in losing some of the secret data. To cope with data loss, data must be embedded in more significant bit planes but it results in lower visual quality of stego image. Hence, it is essential to find ways for adding or subtracting in frequency domain in a way that the inverse transform would ensure integer numbers for the image brightness values in spatial domain; so there will be no need to round them. When the variation size of the brightness intensity in spatial domain is one gray level, the value's effect in the frequency domain should be calculated and the steps in the frequency domain should be adjusted in this accordance. So image mapping from the frequency domain to spatial domain will always produce integer numbers.

As shown in Fig. 4, differential effect of one gray level in spatial domain on the frequency coefficients is equal to a square with the size of the mother wavelet ψ , with centrality of the same pixel. In this case, after obtaining proper frequency coefficients, instead of embedding information just into one frequency coefficient, we can change the coefficient's neighbors

in accordance with the calculated step with ψ . As the result, inverse transform will produce integer numbers for image brightness intensity values and there will be no need for rounding numbers leading to data loss.

For matching the stego image and human visual system more, the proposed algorithm embeds data into image's edges and tries to avoid embedding information into the monotonous parts of the image. Since the average of the window of the mother function ψ is zero, so the average of the coefficients would not be affected by adding or subtracting this window to the center of the frequency coefficients (which represents the edge of the image). We have benefitted this feature in our proposed method and chosen a criteria as the average that is same in both cover and stego images. So the recipient can use the same criteria and estimate absolute correct position of the coefficients containing secret information without a knowledge of the coefficients position in the stego image in advance. It means that, the synchronized receiver can distinguish the location of the information easily.

For choosing edges' coefficients which should carry the information, average of the obtained frequency coefficients should be calculated according to the size of mother wavelet matrix using (5).

$$A_{(u,v)} = \sum_{i=-1}^{i=1} \sum_{j=-1}^{j=1} \frac{1}{9} F_{(u+i,v+j)} \quad (5)$$

where $A_{(u,v)}$ shows the average of the frequency coefficients of F in 3×3 windows. Given the amount of information that is supposed to be hidden in the image, hidden bits per coefficient is chosen in a way that the modified coefficients in the frequency domain do not exceed 70% of the total coefficients. We consider a threshold value and then the candidates for information embedding can be chosen by comparing the threshold value with the average of the coefficients. In this way, the transmitter uses the conditions defined in (6) and subtracts the threshold level with a step of 0.001 unit, as long as number of coefficients is enough for conveying the information.

$$F_{(u,v)} \text{ is selected for embedding if } |A_{(u,v)}| > \text{Threshold} \quad (6)$$

After saving the location of chosen frequency coefficients for carrying information, F should be divided on the central element of mother wavelet matrix, Ψ , and the answer should be rounded. Then the secret data should be embedded in the selected coefficients. In this paper, an algorithm based on [19] is proposed in which the possibility of change in the value of a pixel is less than 0.5. And also our proposed algorithm, in contrast to the algorithm in Jhang and Wang [20], has the ability of embedding multiple bits of data in each coefficient.

If the number of bits embedded in each coefficient is called r , the number of bits carried by two selected coefficients is defined as $s = 2 \times r$. The function value for two selected normalized frequency coefficients can be calculated with (7).

$$f(x_i, x_{i+1}) = [(s-1) \times x_i + s \times x_{i+1} + \text{offset}] \bmod 2^s \quad (7)$$

If the secret information belong to the set of $\{0, 1, \dots, 2^s - 1\}$, we can find x_i and x_{i+1} near the current values that result in same value for f as the value of the secret message. In the information embedding phase, the value of f is calculated for each of the two frequency coefficients. If the calculated f is equal to the secret information, the coefficient will not be changed either. Otherwise, we should seek for x_i and x_{i+1} that have the value for corresponding f equal to the secret message value in their nearest neighborhood (geometrical distance). Then the current frequency coefficient will change into frequency matrix F , by adding or subtracting mother wavelet matrix to or from its center. In this way, the receiver can find out the secret data by calculating f function's value for two newly obtained frequency coefficients.

The offset parameter is added in (7) so the used genetic algorithm can help minimizing changes of the frequency coefficients and consequently the cover image, by optimizing this parameter. Fitness function is Peak Signal to Noise Ratio (PSNR). After information embedding and returning to spatial space, the stego image with the secret information concealed in it will be formed and the fitness value will be calculated for each frequency space using the obtained stego image.

As shown in the flowchart of Fig. 5, crossover and mutation operations are done on the parent chosen from existing chromosomes and then the produced children are assessed. New generation is chosen from current generation and newly produced children with higher fitness. This goes on until the termination condition is met. Convergence of the genetic algorithm results in achieving optimal values for the parameters of the frequency domain and the offset parameter in the information embedding phase. These parameters are considered as a shared key between receiver and transmitter.

2.3. Genetic algorithm

Genetic algorithm is a Search algorithm based on probabilities and can be used to solve various optimization problems [21]. Its theories have been developed by Holland [22] and they are based on natural selection. The main parts of the algorithm can be named as producing initial population, assessment of produced population, parent selection, compounding parent and producing new Offspring, mutations in children and choosing next generation from current population and the children.

Chromosomes generate a population in genetic algorithm and each gene of them is representative of different variants; different chromosomes have different values for each variant. The chromosomes are assessed with a fitness function and regarding natural selection, chromosomes with higher fitness are chosen to last in next generations and produce children. Main goal of the algorithm is finding optimal values of these variables. Each chromosome is composed of some genes which

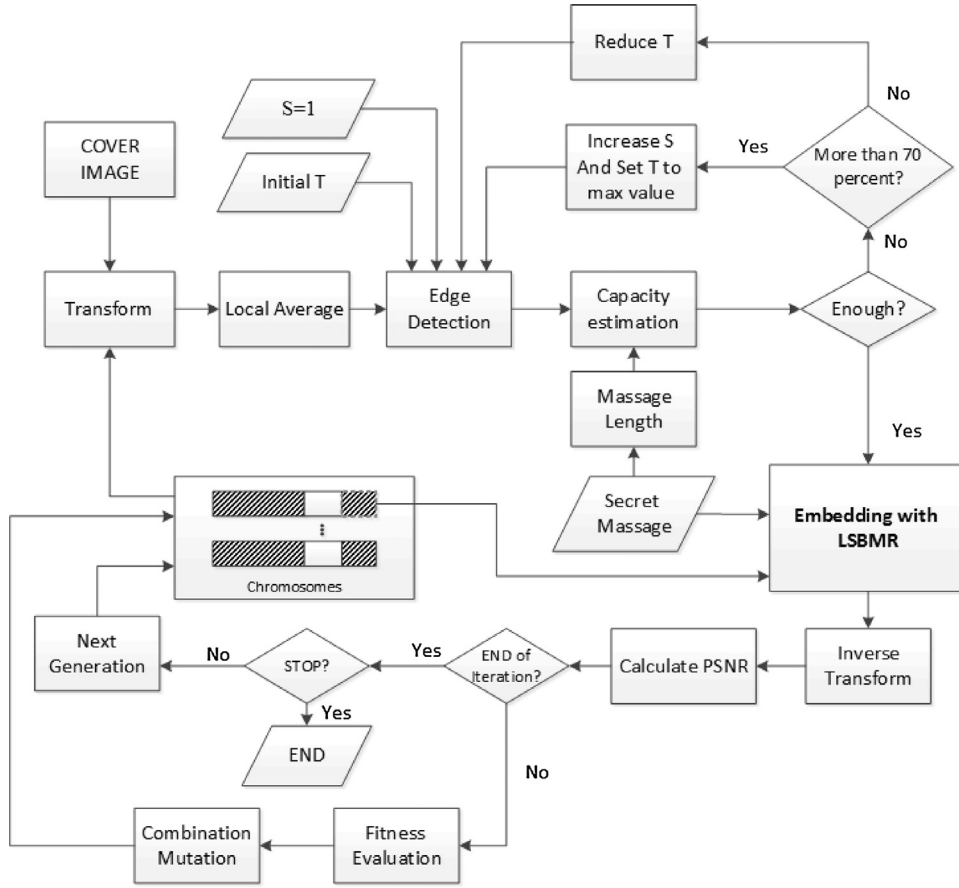


Fig. 5. Block diagram of the proposed algorithm.

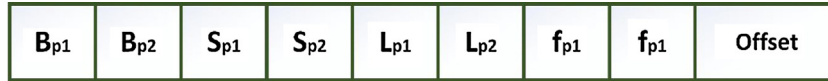


Fig. 6. A sample chromosome used in the proposed algorithm.

are set with the so-called variables. A sample chromosome with the wavelet control parameters used in the proposed algorithm is shown in Fig. 6. Inspired by the natural selection law in nature, parent chromosomes compete with each other and get to be chosen based on their fitness. Those which have the most compatibility with the objective function are chosen with higher probabilities for reproduction according to (8).

$$P_{Ch_i} = \frac{Fitness_{Ch_i}}{\sum_i Fitness_{Ch_i}} \quad (8)$$

in which P_{Ch_i} is the probability of chromosome_{*i*} to be chosen and $Fitness_{Ch_i}$ is the fitness value of chromosome_{*i*}. As in real world, produced child chromosomes inherent from their parents' genes. A sample of child production from two parent chromosomes is shown in (9).

$$\begin{aligned} child_1 &= \alpha P_1 + (1 - \beta)P_2 \\ child_2 &= (1 - \alpha)P_1 + \beta P_2 \end{aligned} \quad (9)$$

in which $child_1$, $child_2$ and P_1 , P_2 are child chromosomes and parent chromosomes respectively. α and β are constant values between 0 and 1. So in child production, there is a probability of producing the next generation from child and parent chromosomes with higher fitness. In this way genetic algorithm finds better values step by step along generations. The algorithm goes on until the termination condition is met.



Fig. 7. Sample images of SIPI data base used as cover images.

Table 1
Parameters of Genetic Algorithm.

Value	Parameter
40	Number of chromosomes
Random (between 0 and 1)	Crossover probability
15%	Mutation probability
16	Number of parents in each iteration

3. Evaluation of the proposed method

In this section simulation results of the proposed algorithm are demonstrated. All the simulations are done with the use of popular images of SIPI [23] data base. Some of the used images are shown in Fig. 7.

All the parameters of the genetic algorithm which was used in the proposed algorithm is inserted in Table 1. All these parameters are gained practically. The population of chromosomes in genetic algorithm are 40 and 16 of them are chosen in each iteration for reproduction of the next generation. Mutation probability in each iteration is adjusted to 15 percent. We have used competitive selection, so the best organism(s) from the current generation are chosen for the next generation. So parent selection in each iteration is done based on Roulette Wheel Selection; so the algorithm can better escape from the local minimums. The principle used for the selection of the next generation is elite selection which helps to speed up the algorithm's convergence.

Termination condition is not static in the proposed genetic algorithm and no more improvements in the fitness value of the chromosomes, for a specific number of iterations, causes the algorithm to stop. The number is chosen 8, so that the searching space will be explored completely. Fig. 8 shows the convergence process of the algorithm for various random data, due to the described termination condition. Fig. 10 shows the results of edge selection particle suggested in our method,

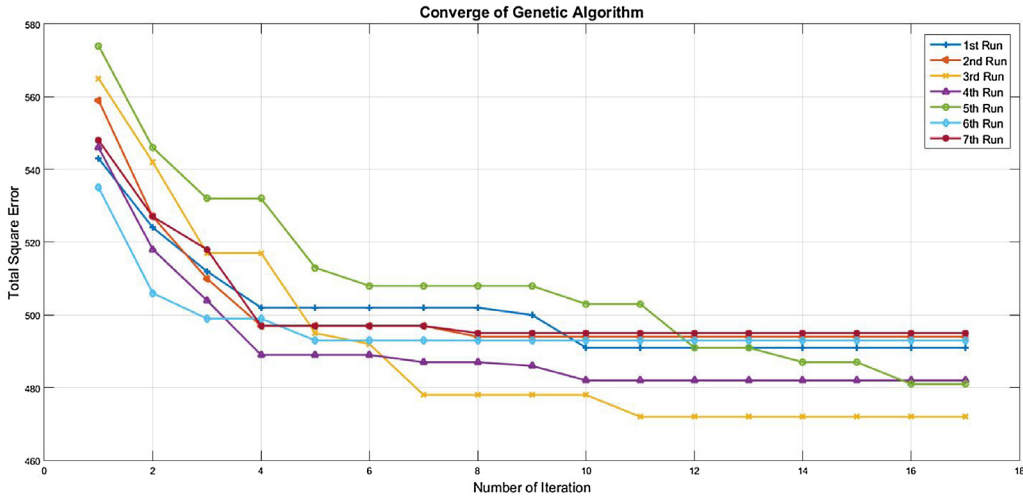


Fig. 8. Convergence process of the algorithm.

for three different volumes of information on Lena, Baboon and F16 images. As it can be seen, in our method, in a total comparative way, edges of the images are selected as the proper areas of the cover image for information embedding.

The value of these chosen areas are absolutely commensurate with the size of the secret data, and if necessary, the value of the suggested points can be increased to ensure that all data is embedded in the image.

Emphasizing on embedding information into the edges of the image arises of the key role of visual system in detecting changes in images. All changes must be such that the visual system cannot detect them.

Another important parameter that was under attention in this article is synchronizing receiver and transmitter and decreasing probability of wrong detection of embedded information locations at the receiver side. This error is different regarding the sensitivity of the edge value calculation approach to the data embedding approach.

Data embedding approaches are usually such that cause trouble in edge points that results in wrong classification of edge points in the receiver side. It is illustrated in Fig. 9 that this error approaches zero with our proposed method. Our method calculates the average value and solves the said problem with benefiting from its characteristic that the average wavelet function is zero.

To compare the proposed algorithm with the previous methods, the value of pixel by pixel difference of the used images is calculated (according to (10)). It is used as cost function for the suggested optimization algorithm.

$$MSE = \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N (I_1(m, n) - I_2(m, n))^2 \quad (10)$$

in which M and N are dimension size of sub images and $I_1(m, n)$ and $I_2(m, n)$ are cover image and stego image respectively.

Another criterion which is used for evaluation is PSNR which is calculated using (11).

$$PSNR(db) = 10 \log_{10} \left(\frac{255^2}{MSE} \right) \quad (11)$$

Since this parameters are based on pixels, it does not matter if the carrier coefficients are edge point or not. Thus, in [24], a criterion is introduced that pays attention to the neighbors of the intended pixel. If the intended pixel is on the edge of the image, it can be changed to somewhat (jnd) and the remaining changes should be considered in calculation of the difference.

Limited amount of the pixel that is allowed to change can be determined according to the pixel value and its surrounding variance. This parameter is considered as PSPNR and is calculated by (12).

$$PSPNR(db) = 10 \log_{10} \left(\frac{255^2}{MSEP} \right) \quad (12)$$

$$MSEP = \frac{1}{MN} \sum_{m=1}^M \sum_{n=1}^N (|I_1(m, n) - I_2(m, n)| - jnd(m, n))^2 \times \delta(m, n) \quad (13)$$

$$\delta(m, n) = \begin{cases} 1 & \text{if } |I_1(m, n) - I_2(m, n)| > jnd(m, n) \\ 0 & \text{if } |I_1(m, n) - I_2(m, n)| < jnd(m, n) \end{cases} \quad (14)$$



Fig. 9. Achieving zero for error detection of information carrier points in the receiver side (a) cover image, (b) chosen positions for information embedding, (c) estimated position in the receiver side.

$$jnd(m, n) = \frac{\delta^2(local)}{5 + I(m, n)} \quad (15)$$

jnd is the threshold value for change proportional to visual system. It specifies the amount of change that a data hiding algorithm can apply on the cover image around the edges of the image.

Simulations are performed with random secret data with a volume of 28800 bits, for three different images from the SIPI database. Simulation results show that obtained stego images are very similar to the original cover image and the difference is not detectable by the human eye. The image difference between the stego image and the cover image (Fig. 10) are the effects of information embedding in frequency domain, leading to changes in the edges of the image. This characteristic of the proposed algorithm ensures high security levels for systems based on it.

The average value of MSE, PSNR, and PPSNR for chosen images from SIPI database with 512×512 size are inserted in Table 2. Our proposed algorithm is compared with the methods in references [17] and [19] which are novel approaches and are used as a part of the proposed method. It can be observed that our proposed method acts better for different volumes of

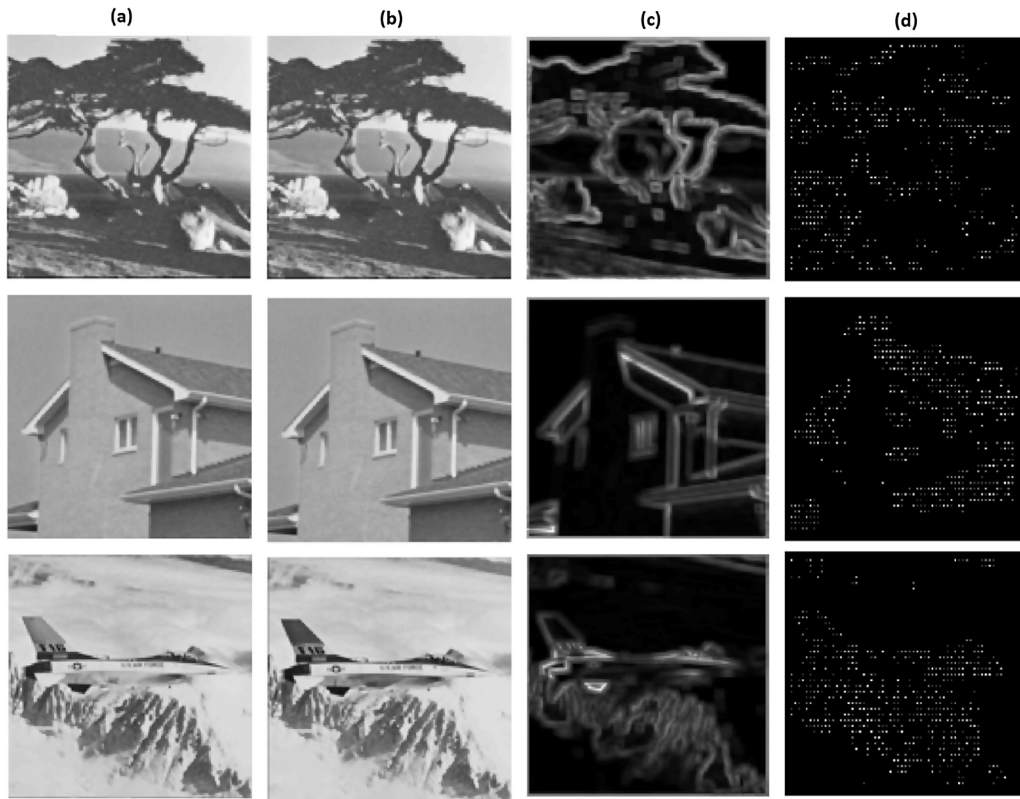


Fig. 10. (a) Cover image, (b) stego image, (c) permitted variations according to visual system, (d) Differences arising on the cover image because of the data embedding.

Table 2
MSE, PSNR, PSPNR for three different algorithms.

Message Size (Kb)	[19]			[17]			Proposed Method		
	MSE	PSNR	PSPNR	MSE	PSNR	PSPNR	MSE	PSNR	PSPNR
6.3	0.0117	67.44	71.81	0.0217	64.76	76.69	0.0081	69.04	80.78
12.8	0.0252	64.11	68.13	0.0460	61.50	71.92	0.0169	65.85	74.20
28.8	0.0787	59.28	62.51	0.1328	56.91	64.45	0.0694	59.71	65.95
51.2	0.1476	56.43	59.88	0.3551	52.62	60.17	0.1270	57.09	61.23
67.7	0.4626	51.47	54.18	0.6094	50.28	54.94	0.4176	51.92	55.07

secret information and ensures 80 dB for the stego image. All the secret information are generated randomly and the results are shown in average. With gradually increasing the volume of secret information, it can be seen that our proposed method degrades in comparison to other methods and it's due to the difficulty of reconciling messages with high-volume with the cover image. PSPNR parameter for the proposed method compared with two other algorithms is shown in Fig. 11.

The method used in [19] is a basic method and part of it is used in our proposed method. This method uses a stream of random numbers for selecting pixels. So it has a good PSNR but doesn't have a high PSPNR. In [17], all data is embedded into the frequency coefficients of an integer wavelet transform and selecting edges is considered in it. So it results in a better PSPNR. Since our proposed method has considered matching the data and the cover image, acts better than two other algorithms and results in a 1.8 dB improvement over two other.

4. Conclusion

In contrast with other papers which select regular frequency domains like integer wavelet transformation in the frequency domain selection phase, in this paper the frequency space is not constant and is chosen according to the secret information and cover image, so the resemblance of secret information and cover image will be maximum. As a result, PSPNR of the cover image will be high after information injection. In this method an extra security layer is added to the algorithm, so the unauthorized receptor, willing to extract secret information, does not know to search information in which frequency space.



Fig. 11. Comparison of PSPNR parameter for the proposed method two other algorithms.

In this paper, the proposed method has shown a 1.8 db improvement over previous works in PSPNR criterion; it shows its better performance.

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